

Valorant Style

1. Game Overview

Hum ne is assignment ke liye ek 5v5 tactical shooter game ko select kiya hai. Is game ka main structure simple hai: do teams hoti hain (Attackers aur Defenders) jo rounds ke hisab se ek dusre ke khilaf khelti hain.

Main Gameplay Mechanics:

- **Precision Shooting:** Shooting isme bohot accurate hoti hai. Har weapon ka apna spray pattern aur recoil hota hai jo real-time mein bullet ki direction decide karta hai.
- **Agent Abilities:** Har character ke paas apni specific powers hoti hain (jaise smoke bombs, flashbangs, ya healing walls) jo map ke gameplay ko dynamic banati hain.
- **Round Economy:** Har naye round ke start par players ko unki performance ke mutabiq match-currency (credits) milti hai, jis se unhein weapons aur abilities buy kar ni hoti hain.

2. High-Level Architecture

Kyunke yeh ek competitive shooter game hai jahan milliseconds ka farq bhi bohot bada hota hai, iske liye hum **Authoritative Server Architecture** use karenge. Iska matlab hai ke poori game ka final control dedicated server ke paas hoga. Client (player ka PC) sirf inputs bhejega aur visual screen render karega, taake lag hacks ya cheating na kar saken.

Major Systems:

- **1. Matchmaking & Connection System**
 - *Simple Explanation:* Yeh system lobby mein players ko unke rank ke hisab se match karta hai aur unhein server se connect karwata hai.
- **2. Combat & Hit-Registration System**
 - *Simple Explanation:* Iska kaam bullet track karna hai. Jab koi fire karta hai, to yeh system check karta hai ke bullet enemy ko lagi ya nahi, aur damage calculate karta hai.
- **3. Round Manager (Game State Control)**
 - *Simple Explanation:* Yeh system match ke phases ko chalta hai. Jaise: Buy Phase (jahan movement lock hoti hai) -> Active Action Phase -> Round Over Phase -> Half-Time Team Switch.
- **4. Economy & Inventory System**

- *Simple Explanation:* Har round ke shuru mein credits distribute karna, items buy karwana, aur jab koi player mare to uska weapon ground par drop karna iska kaam hai.
- **5. Ability Engine**
 - *Simple Explanation:* Characters ki unique abilities ko screen par spawn karna aur unke cooldown timers ko handle karna.

3. Code Structure

Code ko simple aur clean rakhne ke liye hum poore project ko chhote components aur classes mein divide karenge taake hamin ek hi script mein saari logic na likhni pade.

Code Organization Structure:

1 Network:

- NetworkManager karta hai // Server aur client ke beech messages transfer
- LagCompensation adjust karta hai // High ping wale players ke shots code level par

2 Core:

- RoundStateMachine control karta hai // Match ke phases (Buy phase, Active phase) ko

3 Entities

- PlayerCharacter // Har player ka main base container

4 Components

- HealthComponent karta hai // HP, Shields, aur damage calculations handle
- WeaponComponent // Current gun, ammo count, aur fire rate control
- AbilityComponent karta hai // Abilities ke charges aur custom execution handles

Communication Between Systems :

Systems ko aapas mein connect karne ke liye hum Event-Driven Method use karenge taake dependency kam se kam ho.

- **Example :** Jab WeaponComponent server ko batata hai ke fire headshot tha, to server target player ke HealthComponent ki health 0 kar deta hai. Health 0 hote hi ek local event trigger hota hai: OnPlayerDied. Is ek event ke trigger ho baqi saare systems apna apna kaam shuru kar dete hain:
 1. RoundStateMachine check karta hai ke kya poori team khatam ho gayi? Agar haan, to round end kar deta hai.
 2. WeaponComponent player ki gun ko map par drop kar deta hai taake koi aur utha sake.
 3. NetworkManager baqi sab players ko message bhejta hai taake unke UI par kill-feed update ho jaye.

4. Design Decisions

1. Server Authoritative Design with Lag Compensation

- **Reasoning:** Tactical shooters mein cheating sabsay bada masla hoti hai. Agar hum hit-registration client ke PC par chhor denge to log bade aaram se aim-bots bana lenge. Isliye saara decision server karta hai. Rahi baat high-ping players ki, to server back-tracking algorithm use karta hai—yani jab koi shoot karta hai, to server timeline

mein thoda peeche ja kar check karta hai ke player ke screen par enemy sach mein wahan tha ya nahi. Is se game bilkul fair lagti hai.

2. State Machine for Rounds

- **Reasoning:** Agar hum pure game loop ke liye if-else ki lambi conditions lagate to code bohot messy ho jata. State Machine ka faida yeh hai ke jab game BuyPhaseState mein hoti hai, to players ka movement component code-level par disable ho jata hai. Jaise hi timer khatam hota hai, state automatic change ho kar ActionPhaseState par chali jati hai aur movement unlock ho jati hai. Is se logic simple rehti hai aur bugs nahi aate.

3. Component Based Character Setup

- **Reasoning:** Hum ne characters ke liye hardcoded inheritance use nahi ki. Iska faida yeh hai ke kal ko agar hamin koi naya character banana ho jo bilkul alag ho, to hamin naye siray se coding nahi karni padegi. Hum puray bane banaye components (jaise HealthComponent aur WeaponComponent) naye character par attach karenge aur woh ready ho jayega. Code reuse ho jata hai aur project expandable rehta hai.